

Optimization of warfarin dose by population-specific pharmacogenomic algorithm

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To optimize the warfarin dose, a population-specific pharmacogenomic algorithm was developed using multiple linear regression model with vitamin K intake and cytochrome P450 IIC polypeptide9 (CYP2C9*2 and *3), vitamin K epoxide reductase complex 1 (VKORC1*3, *4, D36Y and –1639 G>A) polymorphism profile of subjects who attained therapeutic international normalized ratio as predictors. New algorithm was validated by correlating with Wadelius, International Warfarin Pharmacogenetics Consortium and Gage algorithms; and with the therapeutic dose ($r=0.64$, $P<0.0001$). New algorithm was more accurate (Overall: 0.89 vs 0.51, warfarin resistant: 0.96 vs 0.77 and warfarin sensitive: 0.80 vs 0.24), more sensitive (0.87 vs 0.52) and specific (0.93 vs 0.50) compared with clinical data. It has significantly reduced the rate of over-estimation (0.06 vs 0.50) and underestimation (0.13 vs 0.48). To conclude, this population-specific algorithm has greater clinical utility in optimizing the warfarin dose, thereby decreasing the adverse effects of suboptimal dose. *The Pharmacogenomics Journal* (2012) 12, 306–311; doi:10.1038/tpj.2011.4; published online 1 March 2011

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Introduction

Warfarin is one of the most commonly used oral anticoagulant for the prevention of thrombotic events in subjects with deep vein thrombosis, atrial fibrillation or mechanical valve replacements. Normally, the dose of warfarin will be adjusted by trial-and-error methods or by taking into the consideration of various clinical parameters. Warfarin therapy is often associated with a high risk of major bleeding, especially during the initial phase of warfarin therapy. On the other hand, subtherapeutic dose of warfarin may lead to increased risk of thromboembolic events. Optimization of warfarin dose is challenging because of narrow therapeutic index and greater inter-individual variability in dose requirement.¹ The inter-individual variability in warfarin dose requirement may depend on several factors, for example, dietary intake, variations in pharmacodynamics and pharmacokinetics of warfarin, which include the presence of polymorphisms in cytochrome P450 subfamily IIC polypeptide9 (CYP2C9) and vitamin K epoxide reductase complex 1 (VKORC1) genes.² CYP2C9 is responsible for the metabolic degradation of active warfarin.³ Single-nucleotide polymorphisms in CYP2C9, that is, CYP2C9*2 and CYP2C9*3, decrease the degradation of active warfarin and therefore a lower dose of warfarin is needed to achieve therapeutic international normalized ratio (INR).⁴ VKORC1 is responsible for the activation of vitamin K-dependent clotting factors. By inhibiting VKORC1, warfarin produces anticoagulant effect. VKORC1 –1639 G>A promoter

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polymorphism decreases the function of VKORC1, resulting in lower warfarin requirements in mutant individuals (warfarin sensitivity). VKORC1*3 and VKORC1*4 variants showed intermediate range of warfarin requirement.⁵ On the contrary, a rare Val66Met mutation and D36Y polymorphism in VKORC1 were reported to be associated with warfarin resistance.⁶

To reduce the adverse clinical complications associated with the warfarin over dose or under dose, and to achieve the therapeutic INR at a faster pace, several pharmacogenomic algorithms were developed.^{7,8} In a recent study by Epstein *et al.*⁹ demonstrated the clinical utility of such algorithm in reducing all-cause hospitalization and hospitalization for bleeding or thromboembolism. Majority of algorithms were based on age, height, weight, gender, VKORC1 –1639 G>A, CYP2C9*2 and CYP2C9*3. Certain haplotypes of VKORC1, that is, VKORC1*2 (6853 G>C), VKORC1*3 (9041 G>A), VKORC1*4 (6009 C>T) and VKORC1 D36Y (5417 G>T) that are known to influence warfarin dose,^{5,6} were not included in these algorithms. Vitamin K intake through dietary sources, a likely modulator of INR, was also not incorporated in these algorithms.

Although no such pharmacogenomic algorithm is being followed in India for therapeutic monitoring of warfarin, certain polymorphisms were studied in healthy individuals. A Singapore-based study on Indians demonstrated 18% allele frequency for CYP2C9*3, 12% haplotype frequency for VKORC1 H1 and 82% haplotype frequency for H7, H8 and H9.¹⁰ The allele frequencies for CYP2C9*1, *2 and *3 were 88, 4 and 8%, respectively, among South Indians.¹¹ But no correlation was established between these genetic variants and warfarin dose. The exact incidence of warfarin sensitivity and warfarin resistance in Indian population is not reported yet and also the practicing physicians and surgeons have observed wide inter-individual differences in warfarin dose requirement. In view of these observations, we have developed a new pharmacogenomic algorithm by incorporating additional parameters such as VKORC1 D36Y, VKORC1*3, VKORC1*4 and vitamin K intake in the model to have a more precise prediction of warfarin dose in particular for Indian patients.

Materials and methods

A total of 240 patients who are receiving warfarin were recruited for this study at Nizam's Institute of Medical Sciences, Hyderabad, India. The indications of warfarin prescription were valve replacement ($n=234$), deep vein thrombosis ($n=2$) and atrial fibrillation ($n=2$). The patients who have attained therapeutic INR were grouped as derivation cohort ($n=125$) and the patients who have not attained therapeutic INR were grouped as validation cohort ($n=115$). In derivation cohort, patients who are receiving maintenance dose ≤ 3.0 mg per day were considered warfarin sensitive and patients on warfarin maintenance dose ≥ 10 mg per day were considered warfarin resistant,

while the patients in the range of 3.0–10.0 mg per day were considered to be in the intermediate range.

Clinical information including age, gender, BMI, average daily maintenance dose of warfarin and information about the dietary intake of vitamin K (turnip greens, beetroot, broccoli, lettuce, cabbage, beef liver and so on) were obtained from each patient. Patients with liver, kidney, gastro-intestinal cancer or abnormal bleeding before warfarin therapy were excluded. This study was approved by the Institutional Ethical Committee of the Nizam's Institute of Medical Sciences, Hyderabad. Informed consent was obtained from all the subjects.

Genetic analysis

Blood samples were collected in EDTA. Genomic DNA was isolated using standard protocols. All the DNA samples were tested for CYP2C9*2, CYP2C9*3, VKORC1*3, VKORC1*4, VKORC1 D36Y and VKORC1 –1639 G>A polymorphisms using PCR-restriction fragment length polymorphism approaches. (Table 1) Each PCR reaction was accompanied by a negative control without genomic DNA and a negative control (known heterozygous). The quality of all the restriction enzymes was tested using plasmids with known restriction sites. In $\sim 10\%$ samples, genotyping was repeated to check for genotyping errors and 100% concordance was obtained.

Statistical analysis

Accordance with Hardy–Weinberg equilibrium for polymorphisms was checked using χ^2 -test between observed frequencies and expected frequencies. Using the derivation cohort, stepwise regression was done to derive the pharmacogenomic model for optimizing warfarin dose. The parameters included in this model were age, gender, body mass index, dietary intake of vitamin K, CYP2C9*2, CYP2C9*3, VKORC1*3, VKORC1*4 and VKORC1 –1639 G>A. VKORC1 D36Y was not included in the model as this polymorphism was found to be absent in the study subjects. Drugs interfering with INR were also not being prescribed in any of the study subjects and hence could not be included in the model. To compute genotype data, we used 0, 1 and 2 codes depending on the number of variant alleles. Dietary intake of vitamin K was obtained from the nutritive value of Indian foods. The quantity of each type of food item was multiplied with frequency of consumption. The ratio between the dietary intake of vitamin K by the subject and average intake of vitamin K of the cohort was computed. Multiple linear regression equation was obtained for pharmacogenomic prediction of warfarin dose. Spearman's rank correlation was done to validate the newly developed algorithm with the established algorithms.

The predicted dose of warfarin was correlated with the therapeutic dose by Spearman's rank correlation. In validation cohort, the predicted dose was compared with the clinical dose and INR to assess the accuracy of prediction, sensitivity, specificity, rate of overestimation, rate of underestimation and the applicability in warfarin-sensitive and

Table 1 Reaction conditions for genetic analysis

Polymorphism	Primers	PCR conditions	Restriction enzyme	Restriction site
CYP2C9*2 (Arg144Cys)	5'-TAC AAA TAC AAT GAA AAT ATC ATG-3' 5'-CTA ACA ACC AGA CTC ATA ATG-3'	D: 95 °C/1 min A: 60 °C/30 s E: 72 °C/2 min × 35	Avall	–
CYP2C9*3 (Ile359Leu)	5'-AAT AAT AAT ATG CAC GAG GTC CAG AGA TGC-3' 5'-AAT AAT AAT ATG CAC GAG GTC CAG AGG TAC-3' 5'-GAT ACT ATG AAT TTG GGA CTT C-3'	D: 95 °C/1 min A: 50 °C/1 min E: 72 °C/1 min × 35	NsiI KpnI	– +
VKORC1*3	5'-TTT GCT TTG GCA TGT GAG CCT TGC-3' 5'-ACA GTC CAT GGC AGA CAC ATG GTT-3'	D: 95 °C/1 min A: 64 °C/1 min E: 72 °C/1 min × 30	Acil	+
VKORC1*4	5'-GCA TAA TGA CGG AAT ACA GAG GAG GC-3' 5'-GGT AGA GAC AGG CTT TCA CCA TGT-3'	D: 95 °C/1 min A: 61 °C/1 min E: 72 °C/1 min × 30	Bfal	+
VKORC1 p.D36Y	5'-AAC CTG GAG ATA ATG GGC AGC A-3' 5'-ACA CCG ATC CCA GAC TCCA GAA TA-3'	D: 95 °C/1 min A: 63 °C/1 min E: 72 °C/1 min × 30	RsaI	+
VKORC1 –1639 G>A	5'-ATCCCTCTGGGAAGTCAAGC-3' 5'-CACCTTCAACCTCTCCATCC-3'	D: 95 °C/1 min A: 60 °C/30 s E: 72 °C/2 min × 35	NciI	–

Abbreviations: A, annealing; D, denaturation; E, extension; ×, number of cycles; +, site created due to variant allele; –, site destroyed due to variant allele.

warfarin-resistant cases. Fisher's exact test was used for obtaining all these assay validity parameters.

Results

All the polymorphisms screened were in accordance with the Hardy–Weinberg equilibrium ($P > 0.07$). The variant allele frequencies of polymorphisms in the total cohort were CYP2C9*2, 8.5%; CYP2C9*3, 6.5%; VKORC1*3, 17.0%; VKORC1*4, 7.7%; VKORC1 –1639 G>A, 11.9%. VKORC1 D36Y polymorphism was absent in the study subjects.

In our study subjects, CYP2C9*3, VKORC1*3 and VKORC1 –1639 G>A polymorphisms showed strong association with warfarin sensitivity (<21 mg per week warfarin dose). CYP2C9 *1/*1 (33.39 ± 1.45 mg per week) and *1/*2 (37.84 ± 2.59 mg per week) genotypes showed no statistically significant difference in the warfarin requirement ($P = 0.18$) (Table 2). Males showed higher requirement of warfarin than females ($r = 0.29$, $P = 0.001$). Younger patients showed higher requirement of warfarin than older patients ($r = -0.16$, $P = 0.03$). Body mass index showed no direct influence on warfarin requirement ($P = 0.61$). The influence of vitamin K intake, although not statistically significant ($P = 0.35$), its inclusion in algorithm has improved the overall accuracy of prediction.

By including the physiological parameters namely age, gender, body mass index and dietary intake of vitamin K; and genetic parameters namely CYP2C9*2, CYP2C9*3,

Table 2 The association between polymorphisms and warfarin sensitivity

Polymorphism	Warfarin-sensitive vs other groups	
	Odds ratio	95% confidence interval
CYP2C9*2 (*1/*2)	0.17	0.03–0.99 ^a
CYP2C9*3 (*1/*3)	2.67	1.11–6.48 ^a
VKORC1*3	2.20	1.10–4.44 ^a
VKORC1*4	1.10	0.41–2.93
VKORC1 –1639 G>A	2.35	1.07–5.16 ^a

^a P -value is statistically significant.

VKORC1*3, VKORC1*4 and VKORC1 –1639 G>A in the multiple linear regression model, the following algorithm was obtained from derivation cohort:

Predicted warfarin dose (mg per week) = $-(0.1850797953 \times \text{age}) + (8.107223532 \times \text{male gender}) + (0.1390121899 \times \text{body mass index}) + (6.697769856 \times \text{CYP2C9*2}) - (8.275810018 \times \text{CYP2C9*3}) - (10.72676854 \times \text{VKORC1*3}) + (8.873462677 \times \text{VKORC1*4}) + (11.14958922 \times \text{VKORC1 –1639 G>A}) - (4.824633754 \times \text{vitamin K intake}) + 34.9588747$.

The predicted dose of warfarin with the new algorithm was correlated with Wadelius algorithm ($r = 0.50$, $P < 0.0001$), the International Warfarin Pharmacogenetics Consortium (IWPC) algorithm ($r = 0.31$, $P < 0.005$) and Gage algorithm ($r = 0.23$, $P = 0.004$) (Figure 1).

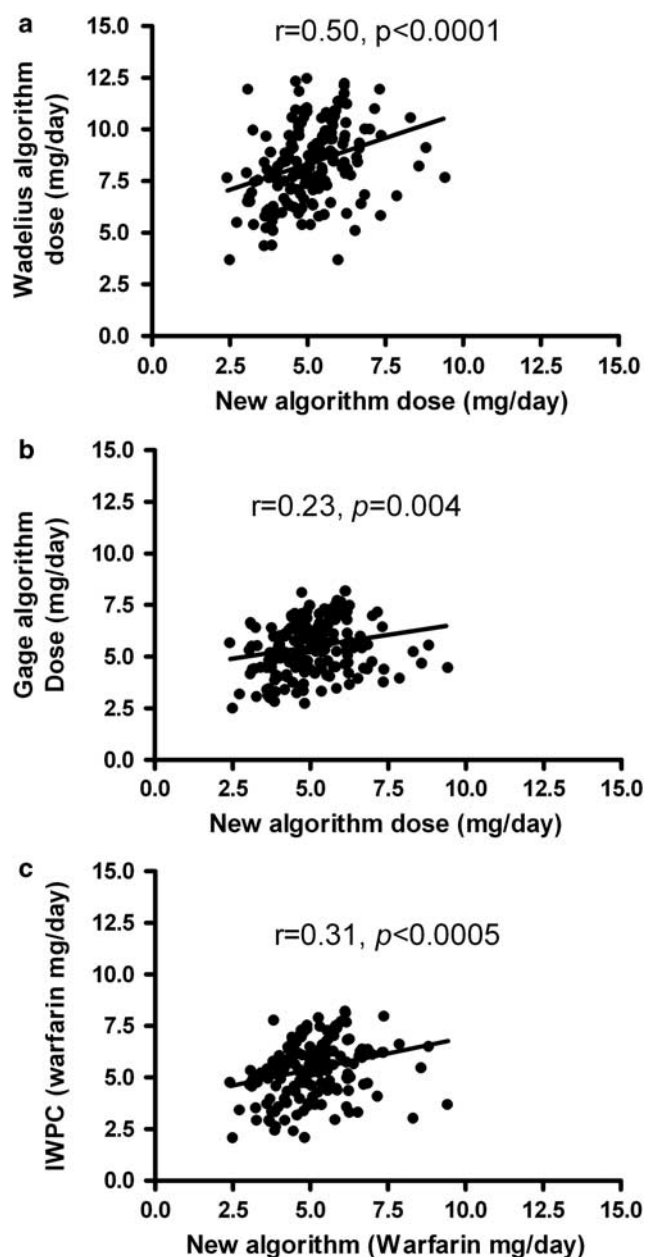


Figure 1 Correlation between predicted doses of warfarin (mg per day) based on new algorithm and (a) Wadelius algorithm, (b) Gage algorithm, (c) International Warfarin Pharmacogenetics Consortium (IWPC) algorithm. Result showed good correlation between predicted doses of new algorithm and Wadelius algorithm ($r=0.50$, $P<0.0001$), Gage algorithm ($r=0.23$, $P=0.004$), IWPC algorithm ($r=0.31$, $P<0.0005$).

The predicted dose of warfarin was correlated with the therapeutic dose ($r=0.64$, $P<0.0001$) (Figure 2). In the validation cohort, this algorithm was accurate in predicting warfarin dose (0.89 vs 0.51) in both warfarin-resistant (0.96 vs 0.77) and warfarin-sensitive (0.80 vs 0.24) groups. This algorithm has increased sensitivity (0.87 vs 0.52) and specificity (0.93 vs 0.50) compared with clinical data.

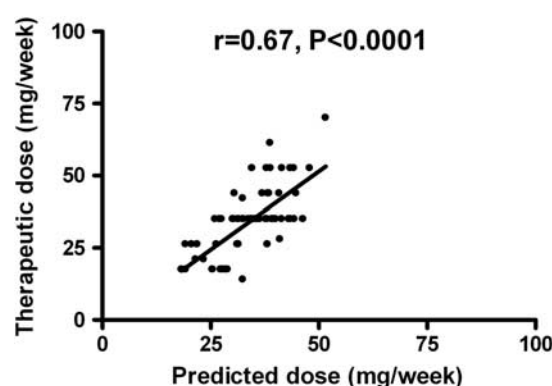


Figure 2 Correlation between predicted dose of warfarin and therapeutic dose of warfarin. Data illustrate that predicted dose of warfarin has good correlation with therapeutic dose ($r=0.64$, $P<0.0001$).

Table 3 Comparison of newly developed algorithm with the clinical parameters

S. No.	Assay parameters	New algorithm (%)	Clinical data (%)
1	Overall accuracy in predicting warfarin dose	89	51
2	Accuracy in predicting warfarin-sensitive group	80	24
3	Accuracy in predicting warfarin-resistant group	96	77
4	Sensitivity	87	52
5	Specificity	93	50
6	Overestimation rate	6	50
7	Underestimation rate	13	48

Significant reduction in the rate of overestimation (0.06 vs 0.50) and underestimation (0.13 vs 0.48) was observed using this algorithm (Table 3).

Discussion

The management of warfarin dosing is challenging due to the narrow therapeutic index and wide inter-individual variability. The inter-individual variability in warfarin dose requirement has been attributed to the several factors such as age, gender, body mass index, diet, disease state, co-medication and polymorphisms in pharmacokinetic and pharmacodynamics pathways of warfarin. Recently, several studies have demonstrated a strong association between the warfarin dose requirements and genetic variations of VKORC1 and CYP2C9.^{2,4-6} To accommodate the influence of all these variables, different pharmacogenomic algorithms were developed for optimizing warfarin dose.^{7,8,12} The versatility of these algorithms was hampered due to potential limitations such as ethnic and population-level differences in the diet and distribution of polymorphisms. The frequencies of CYP2C9 genotypes in different ethnic groups (Whites, Asians and Africans) were *1/*2 (20, 0 and 2%), *1/*3 (12, 4 and 3%), *2/*2 (1, 0 and 0%), *2/*3 (2, 0 and 0%) and *3/*3 (1 and 0%), as observed by Moyer *et al.*¹³

Our study shows 4.25% frequency of CYP2C9 *1/*2, 3.25% frequency of *1/*3 genotype and absence of other genotype combinations such as *2/*2, *2/*3 and *3/*3, which is in agreement with the results demonstrated by Moyer *et al*. The lack of association of CYP2C9 *1/*2 genotype with warfarin sensitivity could be due to no appreciable difference in enzyme activities between *1/*1 and *1/*2 genotypes. Yuen *et al*.¹⁴ have demonstrated differences in the distribution of polymorphisms in pharmacokinetic and pharmacodynamic pathways of warfarin among the three major Asian ethnic groups, that is, Chinese, Malays and Indians in Singapore; however, a very less number of Indian subjects were included in the study. VKORC1 D36Y, which is predominant in Israeli Jews, was totally absent in our study subjects.¹⁵ VKORC1*3 and VKORC1*4 haplotypes that predict intermediate range warfarin requirement were prevalent in our study group. A recent study by Limdi *et al*.¹⁶ also indicated distinct VKORC1 haplotype in Asians compared with other racial groups, specifically in the distribution of -1639 G>A and 1173 C>T single-nucleotide polymorphisms in VKORC1. Among the different algorithms available, only IWPC algorithm has provision to account for unknown polymorphisms in VKORC1; however, the impact of different polymorphisms in this gene might not be the same as used in IWPC algorithm.¹² The dietary differences among different ethnic groups and populations are well established, and among these variables vitamin K intake is crucial that is known to alter INR. To accommodate variables such as VKORC1*3, VKORC1*4 and vitamin K intake for the prediction of warfarin dose, we developed the pharmacogenomic algorithm by incorporating four physiological parameters and five genetic parameters.

Based on the new algorithm, the relative importance of these parameters for predicting the warfarin dose is in the following order: VKORC1 -1639 G>A>VKORC1*3>VKORC1*4>CYP2C9*3>male gender>CYP2C9*2>vitamin K intake>age>body mass index. CYP2C9*3, VKORC1 -1639 G>A and VKORC1*3 polymorphisms were associated with warfarin sensitivity. Among the other variables, male gender and younger age were found to contribute to warfarin resistance to some extent. Body mass index showed no direct influence on warfarin requirement. Kimura *et al*.¹⁷ have also observed the influence of age, gender and body mass index on warfarin dose, with VKORC1 -1639 G>A, CYP2C9*3 and γ -glutamyl carboxylase 8016 G>A accounting for 5.9, 5.2 and 4.6% variance in warfarin dose among Japanese population. The influence of vitamin K intake although not statistically significant, its inclusion in the algorithm was found to improve the overall accuracy in predicting warfarin dose.

To validate the new algorithm, Spearman's rank correlation was done with the three existing algorithms, that is, Wadelius ($r=0.50$, $P<0.0001$), IWPC ($r=0.31$, $P<0.0005$) and Gage ($r=0.24$, $P=0.004$), indicating that all the algorithms showed good correlation with the developed algorithm (Figure 1). The reason for high correlation coefficient of the new algorithm with Wadelius algorithm may be due to the inclusion of similar predictor variables

such as age, gender, VKORC1 -1639 G>A, CYP2C9*2 and CYP2C9*3. The low correlation coefficient with IWPC and Gage algorithms may be due to inclusion of gender, VKORC1*3, VKORC1*4 and vitamin K intake as additional predictors used in the present algorithm. IWPC algorithm was restricted to patients who had a target INR of 2 to 3, whereas the new algorithm covered the patients who had a target INR of up to 3.5. The study group of IWPC algorithm comprising elderly subjects mostly with only 6% of the cohort being younger than 40 years age. In new algorithm, we have included children and younger adults also adequately (65% in derivation cohort and 57% in validation cohort) so that it can be used across a broad range of age.

The predicted dose of warfarin based on new algorithm correlated well with the therapeutic dose ($r=0.64$, $P<0.0001$). By using this algorithm, overall accuracy of prediction was improved significantly (0.89 vs 0.51) in warfarin-resistant (0.96 vs 0.77) and warfarin-sensitive (0.80 vs 0.24) groups. This algorithm has increased sensitivity (0.87 vs 0.52) and specificity (0.93 vs 0.50) compared with clinical data. By using this algorithm, a significant reduction in the rate of overestimation (0.06 vs 0.50) and underestimation (0.13 vs 0.48) was observed.

Our study has certain limitations. The new algorithm performed less predictably among the warfarin-sensitive patients (0.80 vs 0.24). This could be due to unreported polymorphisms or physiological factors that affect metabolic clearance of warfarin. Certain genetic variants such as CYP2C9*8 and CYP4F2 were prevalent among African Americans and Han Chinese populations, respectively.^{18,19} Such variants specific for racial groups might also contribute to the variance in warfarin dose requirement. Factor II Thr165Met and Factor VII G-401T polymorphisms were also reported to be associated with low-dose warfarin requirement.²⁰ Inclusion of lifestyle risk factors such as smoking and alcohol consumption; drugs interfering with INR and polymorphisms in vitamin K-dependent clotting factors as additional predictor variables might help in increasing the precision of prediction. This new algorithm also needs to be tested for its utility in diverse ethnic groups.

To conclude, although the established algorithms are effective in optimizing the warfarin dose, incorporating the additional parameters such as VKORC1*3, VKORC1*4 and vitamin K intake in the algorithm as reported in the current study, will help in precise prediction of therapeutic dose of warfarin. Such precision can be obtained by future studies focused on identifying the new variables that are likely to influence INR.

Conflict of interest

The authors declare no conflict of interest.

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